# PARISHRAM 2.0 2026 (Class 12th)

## PHYSICS

DPP:03

## ELECTRIC CHARGES AND FIELDS

**Q1** If the electric field in a region is uniform and makes an angle of 30° with the surface, and the magnitude of the electric field is 5 N/C and the area is 10 $m^2$ then the electric flux passing through the surface is:

(A) 50 $Nm^2/C$ (B) 43.3 $Nm^2/C$

(C) 100 $Nm^2/C$ (D) 25 $Nm^2/C$

**Q2** Which of the following is NOT a valid application of Gauss's Law?

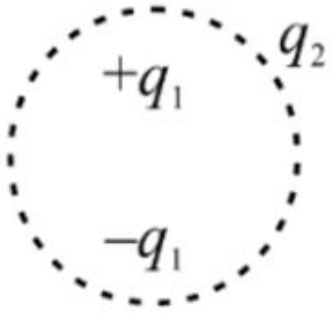
(A) Calculation of the electric field due to a uniformly charged infinite plane sheet.

(B) Calculation of the electric field due to a uniformly charged spherical shell at points inside and outside the shell.

(C) Calculation of the electric field due to a point charge.

(D) Calculation of the electric field due to a charged conductor using Coulomb's law.

**Q3** Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to




(A) q2

(B) Only the positive charges

(C) All the charges

(D) +q1and –q1

**Q4** **Assertion:** The electric flux through a closed surface surrounding a positive charge is proportional to the charge enclosed inside the surface.

**Reason:** According to Gauss's Law, the flux through any closed surface is proportional to the total charge enclosed by the surface and is given by $\Phi_E = \frac{Q_{enclosed}}{\epsilon_0}$

(A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.

(B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.

(C) Assertion is true, but reason is false.

(D) Assertion is false, but reason is true.

**Q5** An electric charge q is placed at the centre of a cube of side a. The electric flux through one of its faces will be

(A) $\frac{q}{6\varepsilon_0}$

(B) $\frac{q}{\varepsilon_0 a^2}$

(C) $\frac{q}{4\pi\varepsilon_0 a^2}$

(D) $\frac{q}{\varepsilon_0}$

**Q6** Total electric flux coming out of a unit positive charge put in air is

(A) $\varepsilon_0$

(B) $\varepsilon_0^{-1}$

(C) $(4\pi\varepsilon_0)^{-1}$

(D) $4\pi\varepsilon_0$

**Q7** The S.I. unit of electric flux is
(A) Weber
(B) Newton per coulomb
(C) Volt × metre
(D) Joule per coulomb

**Q8** A charge q is placed at the centre of the open end of cylindrical vessel. The flux of the electric field through the surface of the vessel is
(A) Zero
(B) $\frac{q}{\varepsilon_0}$
(C) $\frac{q}{2\varepsilon_0}$
(D) $\frac{2q}{\varepsilon_0}$

**Q9** **Assertion:** The electric flux through a spherical surface surrounding a point charge is independent of the size of the sphere.
**Reason:** According to Gauss's Law, the electric flux through a spherical surface only depends on the charge enclosed and not on the radius or size of the sphere.
(A) Both assertion and reason are true, and the reason is the correct explanation of the assertion.
(B) Both assertion and reason are true, but the reason is not the correct explanation of the assertion.
(C) Assertion is true, but reason is false.
(D) Assertion is false, but reason is true.

**Q10** The electric flux through a closed surface area S enclosing charge Q is $\phi$. If the surface area is doubled, then the flux is
(A) $2\phi$
(B) $\frac{\phi}{2}$
(C) $\frac{\phi}{4}$
(D) $\phi$

# Answer Key

| | |
|---|---|
| Q1 | B |
| Q2 | D |
| Q3 | C |
| Q4 | A |
| Q5 | A |
| Q6 | B |
| Q7 | C |
| Q8 | C |
| Q9 | A |
| Q10 | D |

# Hints & Solutions

Note: scan the QR code to watch video solution

**Q1 Text Solution:**

The electric flux is $\Phi_E = E \cdot A \cdot \cos(\theta)$

Substituting the values:

$\Phi_E = 5 \cdot 10 \cdot \cos(30^\circ) = 50 \cdot \frac{\sqrt{3}}{2} = 43.3\ Nm^2/C$

**Video Solution:**

**Q2 Text Solution:**

Gauss's Law can be used to derive the electric field due to a point charge, spherical shell, and infinite plane sheet. However, it cannot directly be applied to charged conductors unless symmetry is considered. Coulomb's Law is typically used to calculate the electric field for such conductors.

**Video Solution:**

**Q3 Text Solution:**

At any point over the spherical Gaussian surface, net electric field is the vector sum of electric fields due to $+q_2$, $q_1$ and $q_2$. Don't confuse with the electric flux which is zero (net) passing over the Gaussian surface as the net charge enclosing the surface is zero.

**Video Solution:**

**Q4 Text Solution:**

The assertion is true because the flux through a closed surface surrounding a charge is proportional to the charge enclosed. The reason is also true, as it is the mathematical statement of Gauss's Law.

**Video Solution:**

**Q5 Text Solution:**

By Gauss's theorem. Total flux $\varphi = \frac{q}{\varepsilon_0}$

So the flux through one face $= \frac{q}{6\varepsilon_0}$

**Video Solution:**

**Q6 Text Solution:**

Total flux coming out form unit charge

$= E.ds = \frac{1}{\varepsilon_0} \times 1 = \varepsilon_0^{-1}$

**Video Solution:**

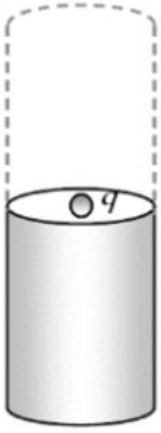
**Q7** **Text Solution:**

S.I. unit of electric flux is $\frac{N \times m^2}{C} = \frac{J \times m}{C} = V \times m.$

**Video Solution:**

**Q8** **Text Solution:**

To apply Gauss's theorem it is essential that charge should be placed inside a closed surface. So imagine another similar cylindrical vessel above it as shown in figure (dotted).




**Video Solution:**

**Q9** **Text Solution:**

The assertion is true because the flux through a spherical surface surrounding a point charge is the same regardless of the size of the sphere. The reason is also true because Gauss's Law states that the flux depends only on the charge enclosed, not on the radius of the sphere.

**Video Solution:**

**Q10** **Text Solution:**

Flux depends on charge enclosed by a close surface.

**Video Solution:**

Android App | iOS App | PW Website